

**EUGLOH: Innovation Days - Immersive technologies bridging art
therapy and mental health**

Technical report

**Tension Resonance: A Projection-Based Art
Therapy Experience for Chronic Musculoskeletal
Pain**



EUGLOH

Team 2

Antonia Kreutzer (Ludwig Maximilian University)

Aloysius Darmawan (Lund University)

Nare Navasardyan (University of Porto)

Swagata Chakraborty (University of Szeged)

Introduction

Chronic pain affects billions of people worldwide and is widely recognized as a major global public health burden due to its high prevalence, associated disability, and impact on quality of life [1]. Among its clinical manifestations, chronic musculoskeletal pain is one of the most prevalent conditions and is characterized by persistent alterations in muscle activity, heightened muscle tension, fear-avoidance behaviors, and disrupted body awareness, all of which contribute to the persistence and amplification of pain over time [2-4]. Existing treatments primarily focus on symptom management, with limited emphasis on self-regulation and embodied interaction [5-7]. Recent research suggests that immersive environments and biofeedback systems can support pain modulation by promoting attentional distraction, relaxation, and reconnection with the body[8-18].

This project presents “Tension Resonance”, a projection-based immersive installation that integrates visuals, sound, movement interaction, and EMG biofeedback to guide users from passive awareness to active physiological control.

System Concept and Design

The system is designed as a multi-stage interactive experience with progressively increasing levels of user engagement. The core concept follows a structured progression from awareness → movement → guided interaction → physiological regulation, enabling users to develop body awareness, engage in controlled movement, and regulate muscle tension through real-time multimodal feedback.

Technical Implementation

The installation is designed to operate across scalable projection configurations (1–4 walls), allowing different levels of spatial immersion. For the prototype, a two-wall configuration was implemented to balance immersive engagement with perceptual clarity and to minimize sensory overload. The experience is structured into four sequential phases, with smooth transitions between each stage. The visual environments were developed using TouchDesigner, while body tracking in the first three phases was enabled using [MediaPipe](#) [19]. A calming soundscape with minimal voice guidance was incorporated to support user focus and maintain a consistent sensory environment.

The first phase introduces a passive entry environment with abstract visuals and ambient audio, allowing minimal interaction through hand tracking. This facilitates initial body awareness without cognitive overload. The second phase expands to full-body interaction, where user movement dynamically modulates visual wave patterns—slow movement produces smooth flows, while rapid movement generates more chaotic responses—encouraging safe exploration

and reducing fear of movement. In the third phase, users engage in guided movement by following a visual model, supporting motor learning, coordination, and confidence in movement. The final phase integrates EMG biofeedback using the MyoWare 2.0 Muscle Sensor in combination with an Arduino microcontroller. Muscle activation is captured in real time and mapped directly to the audiovisual environment: increased tension results in compressed and unstable visual states, while relaxation produces smooth, expansive patterns. This establishes a closed biofeedback loop in which physiological signals are translated into perceptible feedback, allowing users to actively modulate muscle tension through interaction.

Overall, the system aims to enhance body awareness, promote safe and confident movement, reduce muscular tension, and support short-term pain modulation through attentional engagement and multisensory feedback.

Market and Strategy

Based on initial feedback, the project adopts a focused and scalable implementation strategy. The system is intended to be developed in collaboration with a limited number of rehabilitation clinics, working closely with physiotherapists, pain therapists, and patient participants as key stakeholders. This approach enables the identification of practical clinical needs while facilitating the collection of preliminary data to support evidence of effectiveness. Building on this foundation, the system can be iteratively refined and progressively expanded toward broader serviceable and obtainable markets. A primary target region is Europe, where an estimated 100–120 million individuals are affected by chronic pain, with an initial target segment of approximately 1 million potential users.

Discussion and Limitations

The EMG data captured by the system is used as a relative indicator of muscle activity rather than a clinically validated measurement. As such, the system does not provide diagnostic outputs but instead offers real-time, interactive feedback for exploratory and therapeutic purposes. Furthermore, the effects of the intervention are expected to be primarily short-term. Comprehensive validation will therefore require structured user testing and clinical evaluation.

Conclusion

“Tension Resonance” demonstrates the potential of immersive environments to evolve from passive sensory experiences into interactive biofeedback systems. By integrating movement-based interaction with EMG-driven feedback, the installation enables users to develop awareness and actively regulate muscle tension, offering a novel, non-pharmacological approach to pain modulation.

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